

1 IFUL: A Python package for joint-modeling strong 2 gravitational lensing and their source kinematics

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6 Summary

7 IFUL (Integral Field Unit Lensing) is an open-source Python package for joint modeling
8 and simulating strong gravitational lensing systems together with the spatially resolved
9 internal kinematics of background source galaxy. Strong gravitational lensing occurs when
10 the gravitational field of a massive foreground galaxy or galaxy cluster (the “lens”) bends
11 light emitted by a background galaxy (the “source”), magnifying and distorting its appearance
12 into multiple images or arcs. This phenomenon serves as a primary probe in observational
13 astrophysics for measuring cosmological parameters (e.g., the Hubble constant H_0 , the dark
14 energy equation of state w and the matter density of the universe $\Omega_{m,0}$), testing dark matter
15 models, and studying high-redshift galaxy structure and evolution across time.

16 While traditional gravitational lens modeling tools rely primarily on two-dimensional imaging
17 data, IFUL incorporates three-dimensional Integral Field Unit (IFU) spectroscopic datacubes
18 into an end-to-end forward-modeling framework. Integral field spectroscopy captures two-
19 dimensional spatial images where every pixel (spaxel) contains a full spectrum, providing
20 spatially resolved measurements of the internal stellar motion of the galaxy. This allows
21 for our framework to not only flux, but also dynamical information into our model. IFUL
22 forward-models every spaxel in the 3D datacube by unifying macro lens mass profiles with
23 physical models of source galaxy kinematics. By leveraging spatially resolved kinematic markers
24 within lensed arcs and isolating source line emission from foreground lens light, IFUL provides
25 tighter constraints on both lens mass distributions and high-redshift source dynamics than just
26 using the imaging data alone.

27 Statement of need

28 State of the field

29 Software design

30 Research impact statement

31 AI usage disclosure

32 Acknowledgements

33 References